

ShareSphere
CASH-FLOW MINIMISER

HEXA CODERS

PROJECT DOCUMENTATION

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INTRODUCTION

The ShareSphere project is designed to help manage and minimise cash flow among people by calculating net balances and suggesting optimal payments to settle debts. This document provides a detailed project description, including the setup, usage, code explanation, and testing.

WHAT IS A CASH-FLOW MINIMIZER?

A "cash flow minimiser" is a tool or application that helps reduce the number of financial transactions between multiple parties, typically in a group setting where people owe each other money. The goal is to optimise cash flow by simplifying payments ensuring fewer transfers.

For example, if three friends owe each other different amounts of money, instead of each making individual payments, a cash flow minimiser will calculate the simplest way to settle debts. It might suggest that one person pays another directly, eliminating unnecessary intermediate payments and reducing the total number of transactions

PROJECT OVERVIEW

ShareSphere is an innovative fiscal management tool designed to help users effectively manage and minimise their cash flow. Our primary goal is to empower individuals and small businesses to take control of their finances, reduce financial stress, and make informed spending decisions.

CORE FUNCTIONALITY

The project consists of a Python script that calculates the net balances of transactions and suggests optimal payments to minimise cash flow.

USER BENEFITS

By using **ShareSphere**, users can expect to experience a range of benefits, including:

- Reduced financial stress through better visibility of cash flow.
- Improved savings by identifying unnecessary expenses.
- Enhanced financial literacy, equipping users with the knowledge to make smarter financial choices.

TARGET AUDIENCE

ShareSphere is designed for a diverse audience, including individuals, families, and groups, big or small. Whether one is managing household expenses or running a startup, our tool addresses the unique financial challenges faced by each user.

USER EXPERIENCE

We prioritise user-friendliness in our design. The intuitive interface makes it easy for anyone, regardless of their financial background, to navigate the tool and use its features effectively.

Prerequisite Knowledge to Develop ShareSphere

To develop a project like **ShareSphere**, the following foundational knowledge and skills are required:

1. **Python Programming Basics**:
   - Understanding of Python syntax, variables, data types, and basic operations.
2. **Functions**:
   - Knowledge of defining and using functions to modularize code and improve reusability.
3. **Lists and Dictionaries**:
   - Proficiency in managing collections of data using lists and dictionaries.
   - Skills to iterate over these data structures for calculations and updates.
4. **Conditionals and Loops**:
   - Ability to use `if-else` statements and loops (`for`, `while`) for logical flow & repeated tasks.
5. **Error Handling**:
   - Experience with `try-except` blocks to manage invalid inputs and unexpected errors.
6. **Tabulate Library**:
   - Familiarity with using `tabulate` to display data in a structured, tabular format.
7. **Algorithmic Thinking**:
   - Understanding how to design and implement algorithms for tasks like calculating balances and minimizing cash flow.
8. **Input/Output Operations**:
   - Skills to manage user input and display output via the terminal.
9. **Basic Debugging**:
   - Ability to identify and fix errors in Python scripts through debugging tools or manual testing.
10. **Command-Line Operations**:
    - Familiarity with running Python scripts via the terminal or command prompt.

These skills provide a strong foundation for building and expanding the ShareSphere project effectively.

CODE EXPLAINATION

MAIN FUNCTIONALITY

- **`add_users`**: Takes a number of users and their names as input (comma-separated) and adds them to the list of participants.
- **`record_expense`**: Records an expense by specifying the payer, amount, and users involved. Splits the expense evenly and updates balances accordingly.
- **`view_balances`**: Displays the current balances for all participants, showing who owes whom.
- **`settle_debt`**: Records a debt settlement between two users and updates their balances.
- **`main_menu`**: Displays the main menu options for the user and processes their input.

HELPER FUNCTIONS

- **`calculate_net_balances`**: Computes the net balance for each participant by summing up their owed and lent amounts.
- **`minimize_cash_flow`**: Suggests the simplest transactions to settle debts by resolving net balances.
- **`display_balances`**: Formats and displays the balances in a readable tabular form.
- **`get_min_index`**: Identifies the person with the minimum (most negative) balance.
- **`get_max_index`**: Identifies the person with the maximum (most positive) balance.
- **`clear_screen`**: Clears the terminal screen for better readability between operations.
- **`validate_user`**: Checks if a user exists in the system, ensuring inputs are valid.

This modular design ensures the **ShareSphere** application is user-friendly, easy to debug, and efficient for managing group expenses.

TESTING AND VALIDATION

The **ShareSphere Terminal Application** has undergone extensive testing to ensure its reliability and correctness. The tests cover typical use cases and edge cases for all core functionalities.

Key Function Tests

1. `add_users` Function

- Test unique user additions.
- Handle duplicate and empty inputs gracefully.
- Validate handling of whitespace in names.

2. `record_expense` Function

- Test accurate balance updates for valid inputs.
- Handle non-existent users and invalid inputs.
- Reject zero or negative amounts.

3. `view_balances` Function

- Show appropriate messages for no expenses.
- Display correct balances in various scenarios.

4. `settle_debt` Function

- Ensure proper debt settlement logic for valid inputs.
- Handle overpayment and non-existent users gracefully.
- Validate partial debt settlements.

Edge Case and Integration Tests

- Handle invalid names, empty/malformed inputs, and duplicate users.
- Test extreme values (large/small amounts).
- Ensure stability after multiple invalid inputs.
- Validate full workflows: adding users, recording expenses, viewing balances, and settling debts.

This testing ensures the application is robust, handles edge cases, and delivers accurate results.

FUTURE ENHANCEMENTS

1. **Unequal Expense Splitting**: Allow users to split expenses with custom ratios.
2. **User Authentication**: Add login functionality for secure user access.
3. **Expense Categories**: Categorize expenses (e.g., food, transport) for better tracking.
4. **Currency Support**: Enable multi-currency support for international users.
5. **Mobile App Version**: Develop a mobile-friendly version for on-the-go access.
6. **Expense Export:** Export expense reports to CSV or PDF for record-keeping.
7. **Notifications**: Send reminders for unpaid debts or new expenses.
8. **Recurring Expenses**: Allow users to set up recurring monthly expenses.
9. **Debt Summary**: Provide a visual summary or graph of overall group debts.
10. **Group Management**: Enable creation of multiple groups with distinct financial tracking.

Conclusion

The **ShareSphere Terminal Application** is a streamlined and efficient solution for managing group expenses and simplifying financial interactions among participants. By leveraging Python's simplicity and computational power, the application ensures accurate tracking of shared costs, transparent debt calculations, and an intuitive interface for users.

The core functionality, such as adding users, recording expenses, viewing balances, and settling debts, makes ShareSphere ideal for a variety of scenarios, from trips and events to day-to-day expense sharing. With features like balance minimization through the **cash flow optimization algorithm**, it ensures fairness and reduces the complexity of manual calculations. Edge cases and potential user errors have been thoroughly addressed to enhance reliability and user experience.

ShareSphere operates entirely in the terminal, making it lightweight and accessible across different platforms without requiring external dependencies. Its modular structure allows for future enhancements, such as transitioning to a GUI or adding advanced features like currency conversions and expense categories.